

Homework 4

Instructor: Henry Lam

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Problem 1 (Continued from Problem 2 in HW3) A product is classified according to the number of defects it contains and the factory that produces it. Let X_1 and X_2 be the random variables that represent the number of defects per unit (taking on possible values of 0, 1, 2, or 3) and the factory number (taking on possible values 1 or 2), respectively. The entries in the table represent the joint possibility mass function of a randomly chosen product.

$X_1 \backslash X_2$	1	2
0	$\frac{1}{8}$	$\frac{1}{16}$
1	$\frac{1}{16}$	$\frac{1}{16}$
2	$\frac{3}{16}$	$\frac{1}{8}$
3	$\frac{1}{8}$	$\frac{1}{4}$

- a) Find $\text{Var}(X_1)$, $\text{Var}(X_2)$ and $\text{Cov}(X_1, X_2)$.
b) Find the conditional probability mass function of X_2 given $X_1 = 0$.

Solution:

a) We have

$$\mathbb{E}[X_1] = 0 \cdot \frac{3}{16} + 1 \cdot \frac{1}{8} + 2 \cdot \frac{5}{16} + 3 \cdot \frac{3}{8} = \frac{15}{8}.$$

$$\mathbb{E}[X_1^2] = 0^2 \cdot \frac{3}{16} + 1^2 \cdot \frac{1}{8} + 2^2 \cdot \frac{5}{16} + 3^2 \cdot \frac{3}{8} = \frac{19}{4}.$$

$$\text{Var}(X_1) = \mathbb{E}[X_1^2] - (\mathbb{E}[X_1])^2 = 19/4 - (15/8)^2 = 1.2344$$

$$\mathbb{E}[X_2] = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = \frac{3}{2}.$$

$$\mathbb{E}[X_2^2] = 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{2} = \frac{5}{2}.$$

$$\text{Var}(X_2) = \mathbb{E}[X_2^2] - (\mathbb{E}[X_2])^2 = 5/2 - (3/2)^2 = 1/4.$$

$$\mathbb{E}[X_1 X_2] = \sum_{0 \leq i \leq 3, 1 \leq j \leq 2} ij \cdot \mathbb{P}(X_1 = i, X_2 = j) = 1 \cdot \frac{1}{16} + 2 \cdot \frac{4}{16} + 4 \cdot \frac{1}{8} + 3 \cdot \frac{1}{8} + 6 \cdot \frac{1}{4} = \frac{47}{16}$$

$$\text{Cov}(X_1, X_2) = \mathbb{E}[X_1 X_2] - \mathbb{E}[X_1] \mathbb{E}[X_2] = 47/16 - (15/8) \cdot (3/2) = 1/8 \quad (1)$$

b) The conditional probability mass function for X_2 given $X_1 = 0$ is

$$p_{X_2|X_1}(1|0) = \frac{P(X_2 = 1, X_1 = 0)}{P(X_1 = 0)} = \frac{1/8}{1/8 + 1/16} = 2/3. \quad (2)$$

$$p_{X_2|X_1}(2|0) = \frac{P(X_2 = 2, X_1 = 0)}{P(X_1 = 0)} = \frac{1/16}{1/8 + 1/16} = 1/3. \quad (3)$$

Problem 2 If the density function of X equals

$$f(x) = \begin{cases} ce^{-2x}, & \text{for } 0 \leq x < \infty. \\ 0, & \text{for } x < 0. \end{cases} \quad (4)$$

- a) Find the value c .
- b) What is $P\{X > 2\}$?

Solution:

Since the density must integrate to 1:

$$1 = \int_0^{\infty} ce^{-2x} dx = -(c/2)e^{-2x}|_0^{\infty} = c/2. \quad (5)$$

which means $c = 2$. Then $P(X > 2) = \int_2^{\infty} 2e^{-2x} dx = e^{-4}$.

Problem 3 The joint density of X and Y is given by

$$f(x, y) = \begin{cases} xe^{-(x+y)}, & \text{for } x > 0, y > 0. \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

- a) Compute the density of X .
- b) Compute the density of Y .
- c) Are X and Y independent?

Solution:

- a) For $x > 0$, the density $f_X(x)$ is

$$f_X(x) = \int_0^{\infty} xe^{-(x+y)} dy = xe^{-x} \int_0^{\infty} e^{-y} dy = xe^{-x}, \quad \text{and } 0 \text{ for } x \leq 0. \quad (7)$$

- b) For $y > 0$, the density $f_Y(y)$ is

$$f_Y(y) = \int_0^{\infty} xe^{-(x+y)} dx = e^{-y} \int_0^{\infty} xe^{-x} dx = e^{-y}, \quad \text{and } 0 \text{ for } y \leq 0. \quad (8)$$

- c) Yes, since the joint density is equal to the product of the individual densities for any x, y not at the “boundaries” of discontinuity, i.e.,

$$f(x, y) = xe^{-(x+y)} = xe^{-x}e^{-y} = f_X(x)f_Y(y) \text{ for } x > 0, y > 0$$

and $0 = f(x, y) = f_X(x)f_Y(y) = 0 \times 0$ for $x < 0$ or $y < 0$.

Problem 4 The density function of X is given by

$$f(x) = \begin{cases} a + bx^2, & \text{if } 0 \leq x \leq 1. \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

If $\mathbb{E}[X] = 3/5$, find the value of a and b .

Solution:

Since f is a density it integrates to 1 and so

$$1 = \int_0^1 a + bx^2 dx = a + b/3. \quad (10)$$

In addition

$$3/5 = \mathbb{E}[X] = \int_0^1 x(a + bx^2) dx = a/2 + b/4. \quad (11)$$

Solving for a, b , we get $a = 3/5, b = 6/5$.

Problem 5 Suppose that the Rockwell hardness X and abrasion loss Y of a specimen (coded data) have a joint density given by

$$f(x, y) = \begin{cases} 3(x + y), & \text{if } x + y \leq 1, x \geq 0, y \geq 0. \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

- a) Find the marginal densities of X and Y .
- b) Find $\mathbb{E}[X]$ and $\text{Var}(X)$.
- c) Find $\text{Corr}(X, Y)$.
- d) Find the conditional marginal density of X given $Y = 1/2$.

Solution:

- a) The density $f_X(x)$ for $0 < x < 1$ is

$$f_X(x) = \int_0^{1-x} 3(x + y) dy = 3x(1 - x) + 3(1 - x)^2/2 = \frac{3}{2}(1 - x^2) \text{ and } 0 \text{ otherwise.} \quad (13)$$

The density $f_Y(y)$ for $0 < y < 1$ is

$$f_Y(y) = \int_0^{1-y} 3(x + y) dx = 3(1 - y)^2/2 + 3y(1 - y) = \frac{3}{2}(1 - y^2) \text{ and } 0 \text{ otherwise.} \quad (14)$$

- b)

$$\mathbb{E}[X] = \int_0^1 x f_X(x) dx = \int_0^1 \frac{3}{2} x(1 - x^2) dx = 3/8. \quad (15)$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \int_0^1 x^2 f_X(x) dx - (3/8)^2 \\ &= \int_0^1 \frac{3}{2} x^2(1 - x^2) dx - (3/8)^2 = 19/320. \end{aligned} \quad (16)$$

The same goes for $\mathbb{E}[Y]$ and $\text{Var}(Y)$. The answer would be used for part c).

- c)

$$\begin{aligned} \mathbb{E}[XY] &= \int_{x+y \leq 1, x \geq 0, y \geq 0} xy f(x, y) dx dy \\ &= \int_0^1 \int_0^{1-x} 3xy(x + y) dy dx \\ &= 3 \int_0^1 x \left(\frac{x(1-x)^2}{2} + \frac{(1-x)^3}{3} \right) dx \\ &= 1/10. \end{aligned} \quad (17)$$

Then,

$$\text{Corr}(X, Y) = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} = \frac{1/10 - (3/8)^2}{19/320} = -13/19. \quad (18)$$

d) Given $Y = 1/2$, for $0 \leq x \leq 1/2$,

$$f_{X|Y}(x|1/2) = \frac{3(x + 1/2)}{\int_0^{1/2} 3(s + 1/2)ds} = \frac{8}{3}x + \frac{4}{3}, \quad (19)$$

and $f_{X|Y}(x|1/2) = 0$ otherwise.

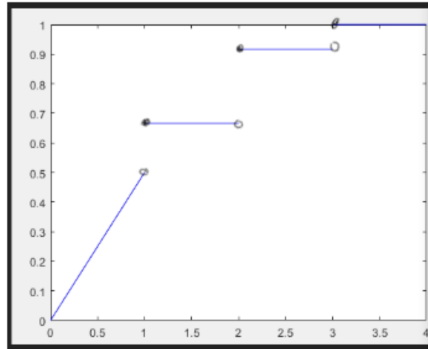
Problem 6 The cumulative distribution function of the random variable X is given as

$$F(x) = \begin{cases} 0, & \text{if } x < 0. \\ x/2, & \text{if } 0 \leq x < 1. \\ 2/3, & \text{if } 1 \leq x < 2. \\ 11/12, & \text{if } 2 \leq x < 3 \\ 1, & \text{if } 3 \leq x \end{cases} \quad (20)$$

- a) Plot the distribution function $F(\cdot)$.
- b) What is $P(X > 1/2)$?
- c) What is $P(2 < X \leq 4)$?
- d) What is $P(X < 3)$?
- e) What is $P(X = 1)$?

Solution:

- a) The cdf plot looks like



As we can see from the graph, the random variable X is has continuous density from 0 to 1 and it has point mass on 1,2 and 3. In particular,

$$\begin{aligned}P(X = 1) &= 2/3 - 1/2 = 1/6 \\P(X = 2) &= 11/12 - 2/3 = 1/4 \\P(X = 3) &= 1 - 11/12 = 1/12.\end{aligned}\tag{21}$$

- b) $P(X > 1/2) = 1 - F(1/2) = 1 - (1/2)/2 = 3/4.$
- c) $P(2 < X \leq 4) = P(X = 3) = 1/12.$
- d) $P(X < 3) = 1 - P(X = 3) = 11/12.$
- e) $P(X = 1) = 2/3 - 1/2 = 1/6.$